

IR Project Proposal

Title:

Hybrid Information Retrieval with Topic Modeling and Knowledge Graph-based Exploration

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Project Description:

This project proposes a hybrid information retrieval system based on recent research in IR, specifically inspired by the paper "Intelligent Scientific Literature Explorer (ISLE)" (2025). The system integrates lexical retrieval using BM25 with semantic retrieval using dense vector embeddings, and combines results through Reciprocal Rank Fusion (RRF) to improve relevance. The retrieved documents are further processed using text clustering techniques such as BERTopic or NMF to generate coherent topics. Additionally, a lightweight knowledge graph is constructed to represent relationships between documents, topics, and metadata. This approach enables both accurate retrieval and structured exploration in an interactive system.